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31N0403

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B.TECH. III SEMESTER MAIN/BACK EXAMINATION 2024-25

COMPUTER SCIENCE AND ENGINEERING

3CS4–03 - Data Structures Algorithms

3AD4-03, 3AM4-03, 3AI4-03, 3CM4-03, 3CY4-03, 3CD4-03, 3CS4-03, 3IT4-03, 3IC4-03 - Common to CS, IT, AI, DS, MC, CM, CD, CA, AD, AM, CY, IO

Time : 3 Hours] **[Max. Marks :** 70

[Min. Passing Marks :

Instructions to Candidates :

Part–A : Short Answer Type Questions (up to **25** words) 10 × 2 = 20 marks. All 10 *questions* are compulsory.

Part–B : *Analytical/Problem* Solving questions 5 × 4 = 20 marks. Candidates have to answer 5 questions out of 7.

Part–C : *Descriptive/Analytical/Problem* Solving questions 3 × 10 marks = 30 marks. Candidates have to answer 3 questions out of 5.

Schematic diagrams must be shown wherever necessary. Any data you feel missing may suitably be assumed and stated clearly. Units of quantities used/ calculated must be stated clearly.

Use of the following supporting materials is permitted during examination. (Mentioned in form no. 205).

1_____ 2_____

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Part-A

Q. 1. What is time complexity of binary search ?

Q. 2. What is the time complexity of following recurrence relation using master method :

$$T(n)=3*T(n/2)+n$$

Q. 3. What is the time complexity of the following code :

```
void fun(int n)
{
for(int i=0;i*i<n;i++)
        printf("Data Structure");
}
```

Q. 4. What is Asymptotic Notation ?

Q. 5. Compute the value of postfix expression using stack : 623 + -382/ + *7*3 +

Q. 6. Compute the postfix expression of following infix expression :

((A-(B+C))*D)*(E+F) ?

Q. 7. What is the suitable and efficient data structure to implement tree ?

Q. 8. What is a Greedy Algorithm ?

Q. 9. What is meant by strongly connected in a graph ?

Q. 10. Given the address of Arr[1000] an array Arr as 10020 and size of each element is 4 bytes in the memory. Find the address of Arr[1700].

Part-B

Q. 1. What is the time complexity of the following recurrence relation :

$$T(n)=T(n/4)+T(n/2)+n*n$$

Q. 2. The Preorder traversal of a tree is : 15, 6, 3, 2, 4, 7, 13, 9, 18, 17, 20. The inorder traversal of same tree is : 2, 3, 4, 6, 7, 9, 13, 15, 17, 18, 20. Construct the tree and write its postorder traversal ?

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Q. 3. Consider a stack which has 2 operation Push and Pop :

Push(a)

Push(b)

Push(c)

ADD

Push(d)

Mul

Push(e)

DIV

Push(f)

SUB

ADD

Find the post order expression equivalent to the above instructions. What is the value of expression if a=2, b=7, c=3, d=4, e=5, f=3 ?

Q. 4. Find the value returned by sample(3) of the below function.

```
void sample(int x) {
static int a;
if(x>0){
sample(x-1);
++a;
samplex(x-1);
printf("%d",a);}
}
```


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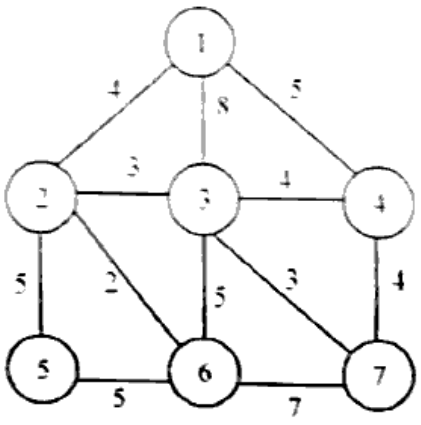
Q. 5. Write a function to add specific node from the list and explain with examples.

Q. 6. Write algorithm of quick sort and compute the time complexity of the quick sort. Apply quick sort for the following list of the elements : 69, 31, 7,18, 21, 44, 5, and 19.

Q. 7. An Array P[30][16] is stored along the row with each of its elements occupying 4 bytes. Find out the base address and address of an element P[2][5], if the location P[5][9] is stored at the address 5000.

Part-C

Q. 1. (i) Find the minimum spanning tree of the graph given below using PRIM Algorithm.



(ii) Demonstrate Dijkstra algorithm on graph shown above where the source vertex is 1. Write the shortest path and its distance from 1 to 7.

Q. 2. What is the output of the below program ? Assume no error in o de and data structure initialized as in figure below.

```
struct node {
    int data;
    struct node *link;
};
void printv(struct *start) {
    struct node *st;
```


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st=start->link;

i. printf("%d ",st->data); // [3 Marks]

start->link=st->link;

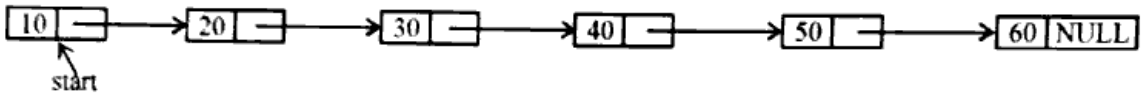
ii. printf("%d ",start->data); // [3 Marks]

st->link->link=st;

iii. printf("%d ",st->link->link->data); // [3 Marks]

iv. printf("%d ",start->link->link->data); // [3 Marks]

v. printf("%d ",sizeof(start->link->link->link)); // [3 Marks]



Q. 3. Construct AVL tree for the following data : Aug, Apr, Jan, Dec, Mar, May, Nov, Jul, Feb, Jun, Oct, Sep. After creating the tree delete following node one by one : Apr, Mar. Follow the order of elements given for insertion and deletion. Show AVL tree after each insertion and deletion. Also mention the type of rotation at each step that is used at the time of insertion and deletion.

Q. 4. Find the value of Toh('S', 'D', 'A', 4).

```
void Toh(char s, char d, char a, int nob)
{
lf(nob>0)
{
    Toh(s,a,d,nob-l);
    printf("%c-----%c",d,s);
    Toh(a,d,s,nob-1);
 }
 }
```


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Q. 5. Consider a stack S with element 0, 1, 3, 4, 5, 6 with top of the stack as 6 and two ques Q1 and Q2. Ql contains elements 1, 2, 3, 8, 9, 0 in the order from front to rear and Q2 is initially empty. What are the contents of Q1 and Q2 if we execute the following code ?

```
void fun()
{
    int x,y;
    while(!empty(S))
{
    x=pop(S);
    y=deque(Q1);
    x=x++ * y++;
    y=x+ ++y;
    x=x++ -y;
    enque(Q1,y);
    enque(Q2,x);
```

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